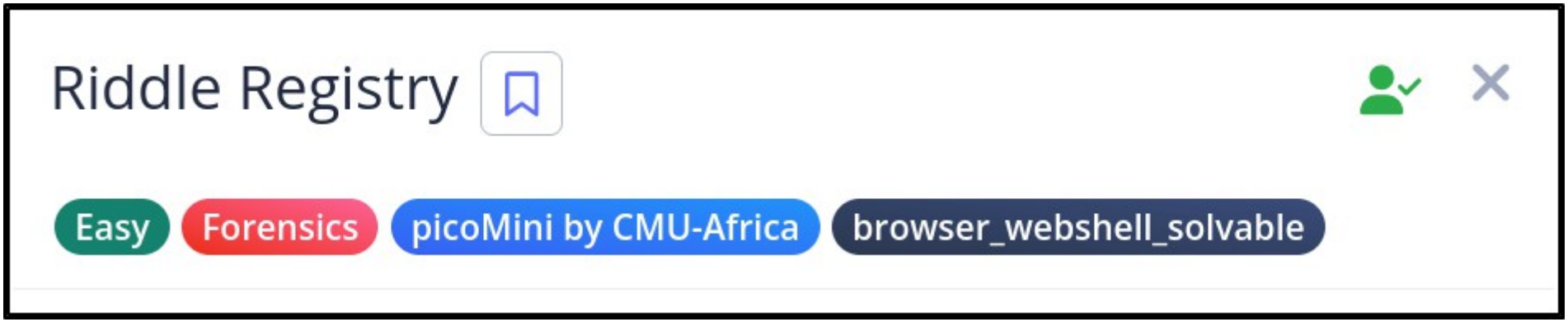
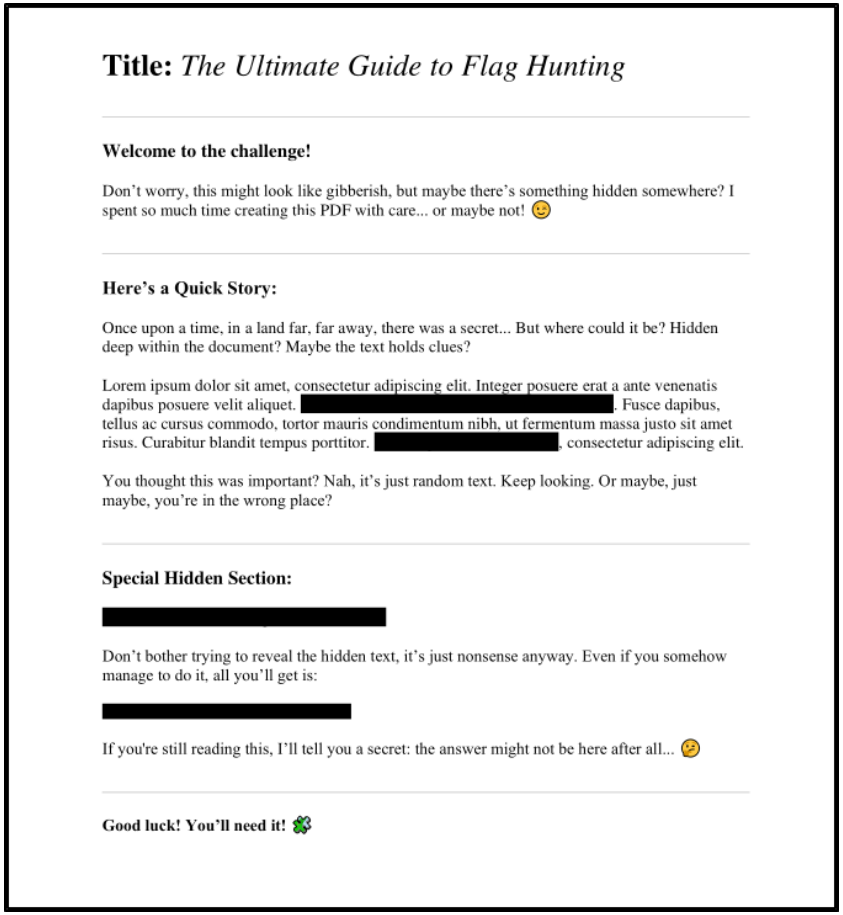
Pico Mini CMU Africa
Riddle Registry



In this challenge, we're asked to find a flag in a PDF document

Pico - Riddle Registry Metadata / EXIF

If we look at the document normally, we don't find the hidden information we're looking for



Title: *The Ultimate Guide to Flag Hunting*

Welcome to the challenge!

Don't worry, this might look like gibberish, but maybe there's something hidden somewhere? I spent so much time creating this PDF with care... or maybe not! 😉

Here's a Quick Story:

Once upon a time, in a land far, far away, there was a secret... But where could it be? Hidden deep within the document? Maybe the text holds clues?

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer posuere erat a ante venenatis dapibus posuere velit aliquet. [redacted]. Fusce dapibus, tellus ac cursus commodo, tortor mauris condimentum nibh, ut fermentum massa justo sit amet risus. Curabitur blandit tempus porttitor. [redacted], consectetur adipiscing elit.

You thought this was important? Nah, it's just random text. Keep looking. Or maybe, just maybe, you're in the wrong place?

Special Hidden Section:

[redacted]

Don't bother trying to reveal the hidden text, it's just nonsense anyway. Even if you somehow manage to do it, all you'll get is:

[redacted]

If you're still reading this, I'll tell you a secret: the answer might not be here after all... 😉

Good luck! You'll need it! 🍀

Pico - Riddle Registry Metadata / EXIF

```
└─$ exiftool confidential.pdf
ExifTool Version Number         : 13.25
File Name                       : confidential.pdf
Directory                       : .
File Size                       : 183 kB
File Modification Date/Time     : 2025:11:07 09:17:39-10:00
File Access Date/Time           : 2026:03:08 13:33:28-10:00
File Inode Change Date/Time     : 2026:03:08 13:32:46-10:00
File Permissions                : -rw-rw-r--
File Type                       : PDF
File Type Extension             : pdf
MIME Type                       : application/pdf
PDF Version                     : 1.7
Linearized                      : No
Page Count                      : 1
Producer                        : PyPDF2
Author                          : cGljb0NURntwdXp6bDNkX20zdGFkYXRhX2YwdW5kIV9jOTk5ZTJhNH0=
```

Most media files (pictures, audio, video) have metadata associated with them, which is data about the data

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File Inode Change Date/Time     : 2026:03:08 13:32:46-10:00
File Permissions                : -rw-rw-r--
File Type                       : PDF
File Type Extension             : pdf
MIME Type                       : application/pdf
PDF Version                     : 1.7
Linearized                      : No
Page Count                      : 1
Producer                        : PyPDF2
Author                          : cGljb0NURntwdXp6bDNkX20zdGFkYXRhX2YwdW5kIV9jOTk5ZTJhNH0=
```

In media files, this metadata is called EXIF, and it can tell us info, such as the device that was used to create the file, the file size, and more

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Metadata / EXIF

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File Inode Change Date/Time     : 2026:03:08 13:32:46-10:00
File Permissions                : -rw-rw-r--
File Type                       : PDF
File Type Extension             : pdf
MIME Type                       : application/pdf
PDF Version                     : 1.7
Linearized                      : No
Page Count                      : 1
Producer                        : PyPDF2
Author                          : cGljb0NURntwdXp6bDNkX20zdGFkYXRhX2YwdW5kIV9jOTk5ZTJhNH0=
```

We can use a common tool in Linux, called Exiftool, to inspect a file's EXIF metadata

Pico - Riddle Registry Metadata / EXIF

```
cGljb0NURntwdXp6bDNkX20zdGFkYXRhX2YwdW5kIV9jOTk5ZTJhNH0=
```

In this case, there's a strange value in the Author field in the EXIF data. This is actually an encoded string, that was encoded using a system called Base64

Pico - Riddle Registry
Base64 Encoding

```
└─$ echo -n "this will be encoded into base64" | base64
dGhpcyB3aWxsIGJlIGVuY29kZWQgaW50byBiYXNlNjQ=
```

```
└─$ echo -n "dGhpcyB3aWxsIGJlIGVuY29kZWQgaW50byBiYXNlNjQ=" | base64 -d
this will be encoded into base64
```

Base64 is a common encoding method found in CTF challenges to disguise information.

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